

Optimization design method for dynamically adjustable variable fractional delay filter

Hai Feng Zheng¹  and Li Ying Han¹

Abstract

When there is a relative difference in the delay scale of different frequency component signals input to a variable fractional delay filter, adjusting a certain sub filter parameter alone cannot flexibly adjust the filter group delay, resulting in distortion errors in the signal spectrum at the output of the filter. An optimization design method for dynamically adjustable variable fractional delay filter is proposed for this purpose. It generates the delay response frequency of an ideal variable fraction filter under the Farrow structure. Based on trapezoidal curves, the multi-scale Fourier spectrum interpolation method is used to linearly combine the sub filters in the Farrow structure to reduce group delay distortion errors. Then, it derives the polynomial parameters of the Farrow structure's sub filters, and optimizes the sub filter parameters uniformly. The method performs real time calculation and updating of adjustable parameters through PID algorithm to optimize and adjust the fractional delay parameters of VFD filters, achieving flexible and adjustable group delay of VFD filters. Through experiments, it has been proven that the VFD filter designed in this paper has low maximum amplitude error, normalized root mean square error, and maximum group delay error. The amplitude frequency response and group delay response can be adjusted flexibly, and it has good application effects.

Keywords

variable fractional delay filter, Farrow structure, FIR filter, transfer function, PID algorithm

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Introduction

Digital filters have the advantages of fast operation speed, good flexibility, high accuracy, strong anti-interference ability, and easy integration. They have been widely used in fields such as communication, speech and image processing, medical and biological signal processing, and spectral analysis.^{1,2} In many signal processing applications, such as digital communication, speech coding and synthesis, sample rate conversion, etc., precise delay processing of signals is required.³ Traditional integer delay filters cannot meet the precise and continuously variable delay requirements, while fractional delay filters can achieve fractional multiple sampling period delays for signals, thus meeting these application requirements. Variable fractional delay (VFD) filter is a type of filter that delays input signals by non-integer multiples of the sampling period, and is considered the first step towards non-uniform sampling discrete-time systems. Unlike traditional filters, VFD filters can flexibly adjust the delay value according to the dynamic changes of the input signal, providing more accurate delay control. They have received widespread attention in modern

engineering application scenarios that require strict delay accuracy and real-time performance. For broadband beamforming technology in fields such as radar and communication, VFD filters can provide more suitable variable fractional delay to address aperture effects in arrays and modify antenna beam pointing. For signal simulators of receiver devices in navigation systems, VFD filters can perform precise delay processing in complex weather environments, thereby improving the performance and accuracy of signal processing.⁴ In digital communication, precise delay control helps improve signal synchronization and anti-interference ability. In speech processing, it can improve the accuracy and clarity of speech recognition. Therefore, studying the optimization of variable fractional delay filters can promote technological progress and application innovation in these fields.

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For the optimization of variable fractional delay filters, Zhou et al.⁵ proposes a new sparse optimization Farrow structure variable fractional delay (SFS-VFD) filter to address the aperture effect in broadband arrays. It is mainly based on coefficient (inverse) symmetry and optimizes the number and order of sub filters, greatly reducing the number of non-zero coefficients. And by using a cost function to solve the parameter minimization problem of multiple regularization constraints, combined with the three block alternating direction multiplier method for solving, the aperture effect can be efficiently corrected, enabling the VFD filter to achieve accurate beam pointing. However, when the three alternating direction multipliers used in this method are used for working conditions with a large number of filter orders or sub filters, they cannot effectively adjust the filter coefficients, resulting in significant filtering delay deviation. Zou et al.⁶ optimizes the symmetry of VFD filter coefficients by optimizing finite impulse response digital filters. By using the traditional least squares method with asymmetric coefficients to optimize the filter and continuing the calculation on half of the tap coefficients of the total order, the computational resources are reduced. However, in the process of optimizing the symmetry of filter coefficients, this method may need to balance the performance of other sub filters. For example, excessive optimization of coefficient symmetry may cause changes in the frequency response characteristics of filters of different orders, thereby affecting the filtering effect and accuracy of the filter. Wang et al.⁷ addresses the issue of requiring a large number of sub filters to be used together in the operation of VFD filters based on the Farrow structure. Through sparse coefficients, the Farrow structure VFD filter is optimized. By using sparse constraint theory, the weight coefficients of the filter are calculated, and the weight coefficient optimization model is improved through L1 regularization under the minimax criterion. The alternating direction multiplier method is further used to iteratively solve the weight coefficients, achieving VFD filter optimization under sparse constraints. However, sparse constraint theory can reduce the number of non-zero coefficients when optimizing filter weight coefficients, but it may also lead to the loss of some important coefficients, thereby affecting the performance of the filter. At the same time, the introduction of L1 regularization enhances the sparsity of the model, but the choice of regularization parameters has a significant impact on the final optimization results. If the parameter selection is improper, it may lead to optimization results deviating from expectations, and even affect the stability and accuracy of the filter. Deng et al.⁸ proposed an optimization method for filter delay. By fitting the time-domain phase of the signal with an exponential polynomial, optimize the filter parameters to reduce delay. However, the optimized filter parameters of this method after exponential polynomial

fitting were not updated in real-time, resulting in insufficient accuracy of the delay response.

Dynamic adjustability refers to the ability of filter parameters (such as time delay, passband width, etc.) to be adjusted in real-time and continuously as needed, without the necessity to redesign the filter. This feature is crucial for signal processing systems as it allows the system to flexibly adapt to changes in input signals under different conditions. This paper dynamically adjusts the delay parameters to optimize the frequency response and phase characteristics of VFD filters, reducing delay errors caused by changes in input signal frequency and phase.

Dynamic adjustable optimization method for variable fractional delay filter

Delay generation of fractional delay filter based on Farrow structure

In wireless communication environments, signals will reach the receiving end through multiple paths. Due to the varying lengths of different paths, the time delay for signals to reach the receiving end is also different, and these delays will vary with the environment, that is, the delay scale. In order to enable the fractional delay filter to flexibly adapt to the time-varying scale relative differences in the fractional delay of different frequency signals, this paper optimizes the design of a variable fractional delay filter using the Farrow structure. After generating the delay signal using the filter, the required fractional delay is obtained through interpolation techniques. The Farrow structure consists of a set of parallel connected fixed coefficient finite impulse response (FIR) sub filters and an adjustable parameter that provides on-site variable delay. By adjusting this adjustable parameter, the time-varying fractional delay scale can be achieved without changing the filter coefficients, thereby improving the flexibility and efficiency of the filter. The designed Farrow structure allows for changing the delay value without recalculating the filter coefficients, thereby improving the real-time performance and flexibility of the system.⁹

The VFD filter designed under the Farrow structure can be transformed into a problem of solving $M + 1$ filter coefficients, which are composed of N -order polynomials of time-delay P .¹⁰ The overall structure of the VFD filter based on the Farrow structure is shown in Figure 1.

As shown in Figure 1, the Farrow structure consists of multiple parallel FIR filters, each sub filter corresponding to one order of polynomial interpolation. Combine the outputs of the sub filters linearly with polynomial weights to generate the final output signal. Receive the score delay value from external input and dynamically adjust the filter output. In the Farrow structure shown in Figure 1, the ideal time delay response frequency of the variable fraction filter is:

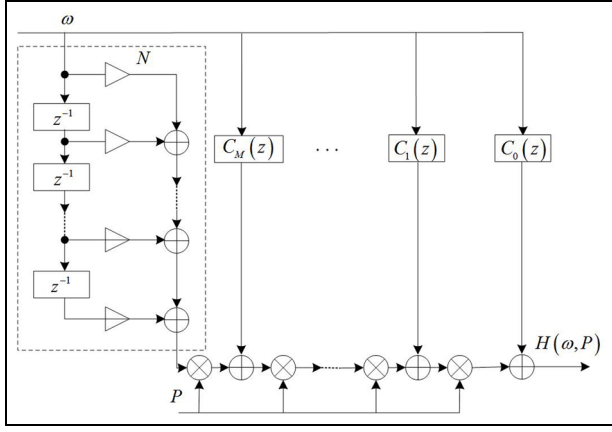


Figure 1. VFD filter structure based on Farrow structure.

$$\begin{cases} H_D(\omega, P) = e^{-\omega P} \\ \quad = \cos(\omega \cdot P) - j\sin(\omega \cdot P) \\ \omega \in [-\omega_P, \omega_P], P \in [-0.5, 0.5] \end{cases} \quad (1)$$

In the formula, ω represents normalized angular frequency; P represents the filter score delay (the delay of non-integer sampling periods introduced by filters in signal processing). Due to the calculation of the ideal delay response frequency of the variable fraction filter and considering the coefficient symmetry of the filter, the delay is a half integer sampling period, therefore the value is $(-0.5, 0.5)$.

Designing an actual filter requires approximating the ideal frequency response through a transfer function, so it is necessary to calculate the transfer function $H(z, P)$ using the following formula:

$$H(z, P) = H_D(\omega, P) \sum_{n=1}^N h_n(P) z^{-n} \quad (2)$$

In the formula, z represents the frequency response parameter; N represents the order of the filter; $h_n(P)$ represents the fractional delay of the filters P at order n , and its calculation formula is:

$$h_n(P) = \sum_{m=0}^M P^m a(n, m) \quad (3)$$

In the formula, m represents the number of filters; $a(n, m)$ stands for sub filter coefficient.

Thus, the transfer function $H(z, P)$ can be expressed as:

$$\begin{cases} H(z, P) = \sum_{n=1}^N \sum_{m=0}^M P^m a(n, m) z^{-n} \\ \quad = \sum_{m=0}^M P^m \times C_M(z) \\ C_M(z) = \sum_{n=1}^N a(n, m) z^{-n} \end{cases} \quad (4)$$

After utilizing the transfer function, we transform formula (4) into the ideal time-delay frequency response $H(\omega, P)$, which is:

$$\begin{cases} H(\omega, P) = \sum_{n=1}^N \sum_{m=0}^M P^m a(n, m) e^{-jn\omega} \\ \quad = \sum_{m=0}^M C_M(\omega) P^m \end{cases} \quad (5)$$

At this point, the delay frequency response $H(\omega, P)$ generated by the filter is the delay generated by a single fractional delay filter under the Farrow structure. However, this result is an ideal result without considering other actual interference factors and needs to be adjusted according to the specific delay requirements of the signal filtering.

Linear combination of Farrow structured sub filters based on multi-scale Fourier spectrum interpolation

The main framework of the VFD filter designed in the previous text is the Farrow structure, which is composed of multiple direct FIR sub filters. When the VFD filter needs to generate a group of multiple frequency components in the signal, there is a delay. When multiple sub filters with nonlinear phase are used for filtering simultaneously, the parameters can be freely adjusted, which will cause multi-scale group delay errors. Therefore, this method is based on the high and low frequency input signals of the filter. The trapezoidal curve is linearly combined with multiple direct FIR filters through multi-scale Fourier spectrum interpolation using normalized convolution windows, thereby solving the problem of free adjustment error of filter group delay.⁴

Using different truncation orders to construct the high and low frequency trapezoidal curve $G_d(\omega, P)$ of the input signal of the filter, the calculation formula is as follows:

$$G_d(\omega, P) = \begin{cases} 1, & \omega/\Delta\omega \in [0, k-1] \cup [S-k+1, S] \\ -\frac{\omega}{\Delta\omega} + k, & \omega/\Delta\omega \in [k-1, k] \\ 0, & \omega/\Delta\omega \in [k, S-k] \\ -\frac{\omega}{\Delta\omega} - S + k, & \omega/\Delta\omega \in [S-k, S-k+1] \end{cases} \quad (6)$$

In the formula, k represents the truncation order; S represents the number of frequency bands; $\Delta\omega = 2\pi/S$ represents transmission bandwidth.

Correspondingly, if the transmission bandwidth $\Delta\omega$ is known, S should be $\lceil 2\pi/\Delta\omega \rceil$, where $\lceil \cdot \rceil$ represents rounding to an integer.

Using the trapezoidal curve $G_d(\omega, P)$ as the transmission characteristic of the FIR filter, assuming its frequency vector is E , and using the Fourier spectrum W_c

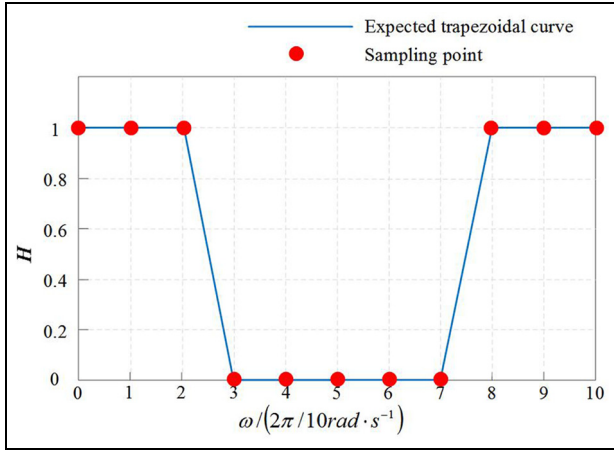


Figure 2. Step curve.

of the normalized convolution window as the interpolation function for $G_d(\omega, P)$:

$$W_c[G_d(\omega, P)] = G_d(\omega, P)\Delta\omega E(k), k = 0, 1, \dots, S-1 \quad (7)$$

Among them, the normalized convolution window is the Hanning window $\Delta\omega$.

Due to the fact that $G_d(\omega, P)$ can pass through all frequency, it approximates a straight line on the transition band $\omega \in [k-1, k]$, which is approximately satisfied at the cutoff frequency ω_{cut} . Therefore, the linear combination of the delay frequency response $H(\omega, P)$ generated by a single filter in the previous section can be expressed as:

$$H = \sum_{k=1}^S W_c[G_d(\omega, P)] + \omega_{cut}H(\omega, P) \quad (8)$$

If the number of frequency bands S and the cutoff frequency ω_{cut} (which can be obtained with the help of a frequency estimator in practical applications) are given, the formula for calculating the truncation order k is:

$$k = \left\lceil \frac{S\omega_{cut}}{2\pi} + \frac{\sqrt{2}}{2} \right\rceil \quad (9)$$

In equation (9), $\lceil \cdot \rceil$ represents rounding to an integer.

The linear combination trapezoidal curve of the Farrow structure sub filter after interpolation processing is shown in Figure 2.

In Figure 2, $S = 10$ and $k = 3$. Due to the zero phase characteristic of H ,¹¹ combined with the domain $n \in [-S+1, S-1]$, the continuous domain $t \in [-S+1, S-1]$ is extended to a discrete domain $n \in [-S+1, S-1]$. As shown in Figure 2, the linear combination of Farrow structured sub filters after interpolation processing has a delay difference of ~ 0 at

both high and low frequencies, effectively reducing the delay deviation of the sub filter group reduces.

Induction and unified optimization of linear combination polynomial parameters for sub filters

Although the Farrow structure's linear combination of sub filters effectively reduced group delay bias in the previous section, it did not take into account that each sub filter in the VFD filter has different filtering weights, and there are differences in the amplitude and frequency of the input signal for each sub filter. If the linear combination interpolation result of filter delay is directly applied, it cannot respond to spectral changes and instead increases delay errors. The delay scale of the VFD filter in the Farrow structure is controlled by the filtering parameters. Therefore, based on the weights, polynomial filtering parameters that integrate sub and uniformly optimized to enhance the flexibility of the delay response and reduce delay deviation. Specifically, in the design process of Farrow structured VFD filters, the first step is to calculate the filtering parameters of each small filter based on the interpolation or extraction ratio.^{12,13} Then, Taylor expansion is performed on the transfer function in two scenarios to derive the sub filter coefficients of the Farrow structure. The sub filter coefficients are unified to complete the sub filter coefficient configuration of the Farrow structure. On the basis of representing the coefficients of the filter in polynomial form with adjustable parameters, the output calculation of the VFD filter is determined. These coefficients determine the weight of each small filter in the filtering process, thereby affecting the final filtering effect. Therefore, the design of VFD filters can be transformed into a problem of optimizing the coefficients of multiple sub filters in the Farrow structure.¹⁴

In order to obtain the filtering parameter $a(n, m)$ of the Farrow structure linear combination sub filter, it is necessary to perform Taylor series expansion on the linear combination filter H . The core of Farrow architecture lies in achieving high-precision signal interpolation.¹⁵ Through Taylor series expansion, the transfer function can be expressed as the sum of a series of polynomial terms that can accurately approximate the characteristics of the original signal near the interpolation point. This approximation method has high accuracy, especially near the interpolation points, thus ensuring that the Farrow structure maintains high signal quality during the interpolation process.¹⁶ Through Taylor series expansion, complex transfer functions can be decomposed into a series of simple polynomial terms. These polynomial terms can be directly used for sub filter design in Farrow structures, simplifying the filter design process. In addition, due to the local approximation property of Taylor series expansion, the appropriate order can be dynamically selected according to actual needs to balance computational

complexity and approximation accuracy, further simplifying the implementation of filters.¹⁷

Specifically, excluding the boundary points $-S + 1$ and $S - 1$ in the discrete domain $n \in [-S + 1, S - 1]$, the fixed points for Taylor expansion can be selected as the remaining $2S - 3$ continuous values $t = n$ of the all phase filter (a filter that superimposes and filters all phase components of an input signal), with $n \in [-S + 2, S - 2]$. Without loss of generality, let $t = n + P$ and discuss it in two situations.¹⁸

1. *Scenario 1:* $0 \leq P \leq 0.5$, in which case t falls within the interval $[n, n + 1]$, therefore H can be approximated by the Taylor coefficient expansion of $H_n(z, P)$. The calculation formula is as follows:

$$\begin{aligned} H &\approx H_n(n + P) \\ &= H_n(n) + g_{1,n}(n)P + \frac{3g_{2,n}(n)P^2 + 2g_{3,n}(n)P^3}{6} \end{aligned} \quad (10)$$

Base on formula (10), the form of the sub filter parameter $a(n, m)$ of the Farrow structure can be derived, which can be expressed as:

$$a(n, m) = \frac{H_n(n + P)g_{mn}(n)}{m}, m = 0, 1, 2, 3 \quad (11)$$

Furthermore, the expression of the derivatives of each order in formula (11) can be derived as follows:

$$\left. \begin{aligned} g_n(n) &= g(n), m = 0 \\ g_n^{(1)}(n) &= b_{n,1} + 2b_{n,2}n + 3b_{n,3}n^2, m = 1 \\ g_n^{(2)}(n) &= 2b_{n,2} + 6b_{n,3}n, m = 2 \\ g_n^{(3)}(n) &= 6b_{n,3}, m = 3 \end{aligned} \right\} \quad (12)$$

By combining formulas (11) and (12), the coefficients of the sub filters can be summarized and unified. The calculation formula is:

$$a(n, m) = \begin{cases} \sum_{i=0}^3 b_{n,i}n^i, & m = 0 \\ \sum_{i=m}^3 b_{n,i} \prod_{j=3-i}^{m+2-i} (3-j)n^{i-m}, & n \in [-S + 2, S - 2] \\ \frac{\sum_{i=m}^3 b_{n,i} \prod_{j=3-i}^{m+2-i} (3-j)n^{i-m}}{m}, & m = 1, 2, 3 \end{cases} \quad (13)$$

2. *Scenario 2:* $-0.5 \leq P \leq 0$. In this scenario, t falls within $[n - 1, n]$, and H can be approximated by the Taylor coefficient expansion of $H_{n-1}(z, P)$. The calculation formula is as follows:

$$\begin{aligned} H &\approx H_{n-1}(n + P) = H_{n-1}(n) + g_{1,n-1}(n)P \\ &\quad + \frac{3g_{2,n-1}(n)P^2 + 2g_{3,n-1}(n)P^3}{6} \end{aligned} \quad (14)$$

According to the same logic as in situation 1, the polynomial parameters of the sub filter can be summarized and optimized uniformly. The calculation formula is:

$$a(n, m) = \begin{cases} \sum_{i=0}^3 Hb_{n-1,i}n^i, & m = 0 \\ \sum_{i=m}^3 b_{n-1,i} \prod_{j=3-i}^{m+2-i} (3-j)n^{i-m}, & n \in [-S + 2, S - 2] \\ \frac{\sum_{i=m}^3 b_{n-1,i} \prod_{j=3-i}^{m+2-i} (3-j)n^{i-m}}{m}, & m = 1, 2, 3 \end{cases} \quad (15)$$

Dynamic adjustable control of polynomial parameters for VFD filter based on PID algorithm

As can be seen from the previous section, the control mechanism of adjustable parameter P is crucial when optimizing the polynomial parameter $a(n, m)$ of VFD filters. This mechanism enables real-time adjustment of filter delay according to actual needs, thereby achieving precise control of signal phase and delay.¹⁹ When P changes, the output signal of the filter will correspondingly undergo phase shift and delay adjustment. In order to achieve dynamic adjustment of VFD filters, PID algorithm is used to calculate and update adjustable parameters in real time. The core idea of PID controller is to change the control quantity based on the error, thereby achieving the effect of reducing the delay error. In the delay control of filters, the error between the actual delay and the target delay is the key to adjusting the control parameters. By using a PID controller, this error can be monitored in real-time and adjustable parameters can be adjusted based on the magnitude and direction of the error, thereby achieving precise control of filter delay.

On the basis of representing the coefficients of the filter in polynomial form with adjustable parameters, the output calculation of the VFD filter is determined using the following formula:

$$P(t) = \sum_{n=0}^{N-1} P_n x(t - n) \quad (16)$$

In the formula, $P(t)$ represents the output signal of the filter; $x(t)$ represents the input signal; P_n represents the n th dynamically adjustable parameter of the filter.

Using a PID controller to update adjustable parameters in real-time, the formula is as follows:

$$P(t + 1) = P(t) + K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt} \quad (17)$$

In the formula, $e(t)$ represents the error signal (i.e. the difference between the actual delay and the target delay); K_p , K_i , and K_d represent proportional, integral, and differential gains, respectively.

The steps for implementing dynamic adjustability of VFD filters through PID algorithm are as follows:

1. *Determine filter parameters:* Select appropriate filter order N , polynomial order M , and PID controller parameters according to application requirements. Select the filter order based on the frequency domain transition band width, time domain constraints, and signal bandwidth standards. Select polynomial order based on delay accuracy requirements, computational complexity, and signal frequency range standards.
2. *Initialize filter coefficients:* Perform Taylor series expansion on the transfer function.
3. *Real time calculation error:* During the operation of the filter, the error between the actual delay and the target delay is calculated in real time.
4. *Update adjustable parameter P :* Use PID controller to update formula (17) to adjust adjustable parameters in real time.
5. *Update filter coefficients:* Substitute the new adjustable parameter value P into formula (11) to re-optimize the sub filter coefficients $a(n, m)$ of the Farrow structure.
6. *Output filter signal:* Use the updated filter coefficients to calculate and output the signal of the VFD filter.

By designing a reasonable control mechanism for adjustable parameter value P based on the above algorithm, the dynamic adjustability of VFD filters can be achieved. This mechanism not only improves the flexibility and adaptability of the filter, but also provides more accurate and reliable signal processing methods for practical applications. Thus, the optimization design of a dynamically adjustable variable fractional delay filter has been completed. The overall design process is shown in Figure 3.

Experimental analysis

Experimental setup

To verify the application performance of the dynamically adjustable VFD filter designed in this paper, a simulation example is designed for comparative verification. Compare the performance of the VFD filter designed in this paper with the VFD filter designed based on least squares method (Wang et al.⁷) and the VFD filter designed based on sparse constraints (Zou et al.⁶).

In this paper we use FPGA integrated chip data as the input signal source for the filter, and the experimental scenario is shown in Figure 4.

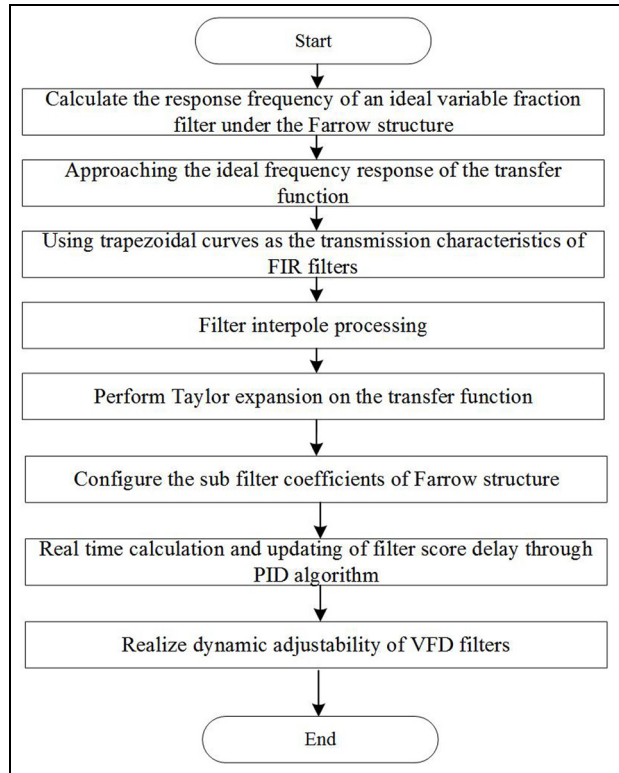


Figure 3. Optimization flowchart of variable fractional delay filter.

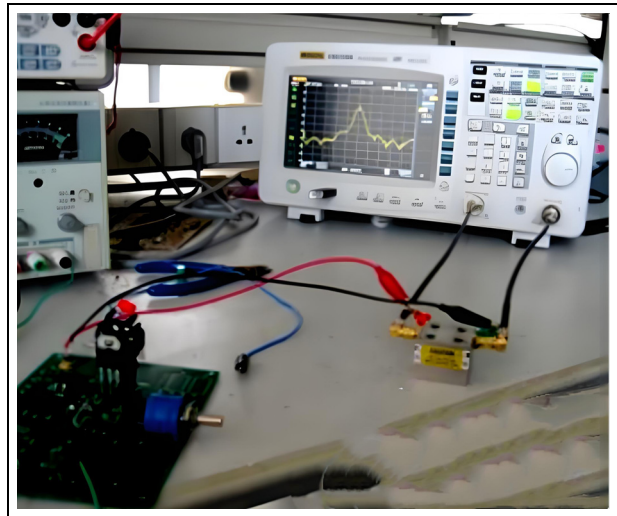


Figure 4. Experimental scenario of FPGA integrated chip filtering.

The simulation experiment of Farrow filter was conducted using Matlab + Modelsim joint simulation method. The testing and verification mainly include the generation of test data, sub filter coefficients, digital prototype (STM 32F103CBT6), excitation file test-bench (MTC – 3000), Farrow filter, etc. The testing and verification process is shown in Figure 5.

The introduction of each module is as follows: the test data generation module is used to generate the

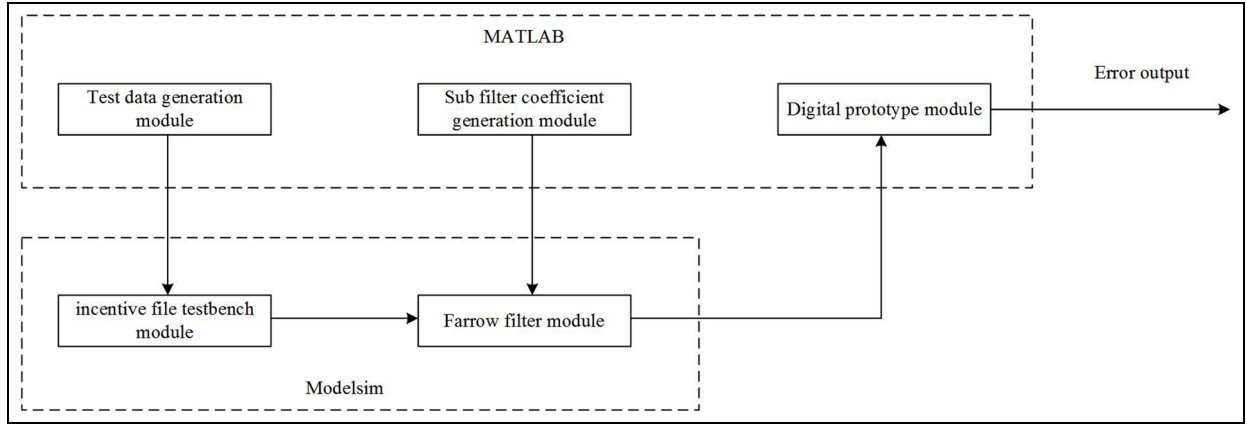


Figure 5. Testing and verification process.

required test data; the sub filter coefficient generation module is used to generate Farrow filter parameters; the digital prototype module is used to calculate theoretical filtering results and automatically compare them with simulation results; the incentive file testbench module is used to generate driver files for Modelsim simulation; the Farrow filter module is the FPGA engineering file to be tested.

Set the order N of the filter to 25, the number M of sub filters to 3, 5, 7, and 9, the cutoff frequency $\omega_{cut} = 0.6\pi$, the penalty factor $\lambda = 0.1$ (constraint strength used to quantify the output of the filter), $p = [-0.5, 0.5]$, and the zero threshold $\gamma = 10^{-6}$ (set to zero when the absolute value of the filter output is too small).

Experimental results and analysis

Test of delay generation effect of fractional delay filter. The input test data for the FPGA integrated chip is a set of sine wave signals, and the filter parameters are generated using the Lagrange interpolation method. The delay results of the Farrow structure filter group generated in this paper are plotted using Matlab, as shown in Figure 6.

As shown in Figure 6, this paper uses multiple sub filters with Farrow structure to effectively generate filtering delay results for different frequency and amplitude signals with delay scales d of 0–0.9, without delay overlap, and the delay generation effect is good.

Unified optimization effect of linear combination polynomial parameters induction for sub filters. Use the maximum amplitude error $\varepsilon_{\max}$, normalized root mean square error e_2 , and maximum group delay error $\varepsilon_{p\max}$ to evaluate the optimization effect of the sub filter linear combination polynomial parameters of the filter.

1. The maximum amplitude error refers to the maximum difference between the output signal of the filter and the expected output signal,

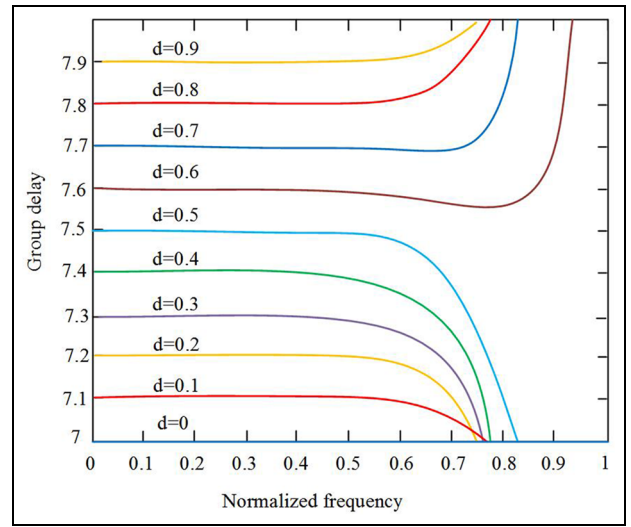


Figure 6. Farrow structure filter group delay.

which reflects the maximum degree of distortion that the filter may produce when processing the signal. By evaluating the maximum amplitude error, it can be ensured that the filter will not produce excessive delay distortion when optimizing the parameters of the sub filter linear combination polynomial, thereby meeting the performance requirements of the system. The calculation formula is as follows:

$$\varepsilon_{\max} = \max\{20 \lg |eH(\omega, P)|\} \quad (18)$$

2. Normalized root mean square error is a statistical indicator used to measure the difference between the output signal of a filter and the expected output signal. It can be obtained by calculating the square root of the average sum of squares of the difference between the output signal of the filter and the expected output signal, and normalizing it. MSE has a high sensitivity to errors, which can

Table 1. Comparison of error accuracy of different filters (dB).

Linear combination sub filter filtering parameters $a(n, m)$	Proposed method			Based on the least squares method ⁷			Based on sparse constraints ⁶		
	$\varepsilon_{\max}$	e_2	$\varepsilon_{p \max}$	$\varepsilon_{\max}$	e_2	$\varepsilon_{p \max}$	$\varepsilon_{\max}$	e_2	$\varepsilon_{p \max}$
3	-17.89	0.0068	0.0036	-18.45	0.0069	0.0039	-19.56	0.0084	0.0043
5	-19.18	0.0063	0.0031	-20.03	0.0070	0.0036	-20.67	0.0078	0.0038
7	-20.13	0.0057	0.0028	-20.79	0.0068	0.0029	-20.45	0.0076	0.0036
9	-20.45	0.0057	0.0028	-20.99	0.0068	0.0029	-20.35	0.0077	0.0035

comprehensively reflect the performance of the filter in the entire signal processing process, and also reflect the small distortion of the filter in signal processing. The calculation formula is as follows:

$$e_2 = \frac{\int_0^{a\pi} \int_{-0.5}^{0.5} |eH(\omega, P)|^2 dP d\omega}{\int_0^{a\pi} \int_{-0.5}^{0.5} |H_d(\omega, P)|^2 dP d\omega} \times 100\% \quad (19)$$

3. The maximum group delay error refers to the difference between the maximum group delay generated by the filter during signal processing and the expected group delay. By evaluating the maximum group delay error, it can ensure that the filter does not produce excessive phase distortion during signal processing, thereby meeting the phase response requirements of the system. The calculation formula is as follows:

$$\varepsilon_{p \max} = \max\{|\tau(\omega, P) - P|\} \quad (20)$$

The performance comparison of VFD filters designed by three algorithms, $\varepsilon_{\max}$, e_2 , and $\varepsilon_{p \max}$, is shown in Table 1.

Based on Table 1, it can be found that when the filtering parameters $a(n, m)$ of the linear combination sub filter are 3, 5, 7, and 9, and $N = 25$, the maximum amplitude errors of the VFD filter are -17.89, -19.18, -20.13, and -20.45 dB, respectively, all of which are smaller than the maximum amplitude errors of the VFD filters of the other two compared methods. The normalized root mean square errors of the VFD filter designed in this paper are 0.0068, 0.0063, 0.0057, and 0.0057, respectively, which are still smaller than the normalized root mean square errors of the VFD filters of the other two methods compared. The maximum group delay errors of the VFD filter designed in this paper are 0.0036, 0.0031, 0.0028, and 0.0028, respectively. The results are the same as the maximum amplitude error and normalized root mean square error, both of which are smaller than the maximum group delay errors of the VFD filters compared to the other two methods. This verifies the accuracy of the algorithm proposed in this paper and demonstrates that the VFD filter designed in this paper has high accuracy in application. This is because the VFD filter designed in

this paper summarizes and optimizes the linear combination polynomial parameters of the sub filter through Taylor series expansion, accurately approximating the characteristics of the original signal near the interpolation point and directly used in the design of sub filters in the Farrow structure. It can dynamically select the appropriate order to balance computational complexity and approximation accuracy according to actual needs.

Dynamic response adjustable control effect of VFD filter polynomial parameters. We use amplitude frequency response and group delay response, respectively, to verify the dynamic response adjustable control capability of the filter.

1. The amplitude frequency response describes the attenuation or gain of signal amplitude by a filter at different frequencies, and can intuitively reflect the processing effect of the filter on signal frequency components. By testing the amplitude frequency response, we can understand the degree of attenuation of harmonics and noise by the filter at different frequencies, thereby verifying its adjustable filtering delay effect.
2. Group delay refers to the relative delay difference introduced by a filter to signals of different frequency components, reflecting the distortion of the signal spectrum by the filter. Especially for nonlinear phase filters, the group delay effect is more pronounced. The testing of group delay response can evaluate the impact of filters on signal delay, thereby understanding their performance in practical applications. The group delay response test helps ensure that the VFD filter does not introduce excessive delay while filtering out harmonics and noise, thereby maintaining the real-time and accuracy of the signal.

The comparison of amplitude frequency response and group delay response of different methods is shown in Figures 7 and 8, respectively.

From the comprehensive analysis of Figures 7 and 8, it can be seen that the method proposed in this paper has a flatter curve at low frequencies, performs better than the VFD filter designed based on the least squares method at high frequencies, and performs better than

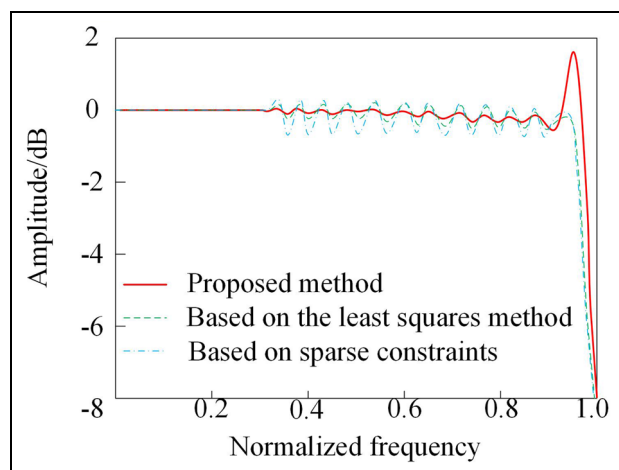


Figure 7. Comparison of amplitude frequency response of different methods.

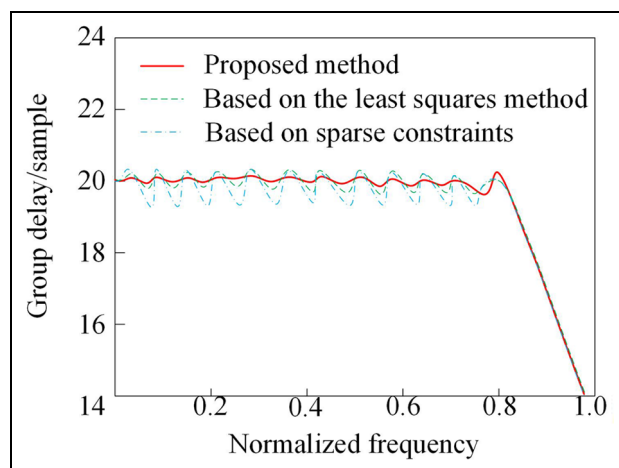


Figure 8. Comparison of group delay response using different methods.

the VFD filter designed based on sparse constraints at all frequency bands. This is because the VFD filter designed in this paper uses a PID controller to update adjustable parameters based on the frequency band of the input signal, thereby achieving free adjustment of the filter group delay, enabling the delay results to maintain high accuracy at both high and low frequencies.

Conclusion

VFD filters play an important role in the field of signal processing, and their accuracy and response speed have a significant impact on the overall performance of the system. With the continuous development of technology, the performance requirements for VFD filters are also constantly increasing. Therefore, this paper focuses on optimizing the dynamic adjustable performance of variable fractional delay filters. In experimental verification, it was found that the designed VFD filter

exhibited low error levels in terms of maximum amplitude error, normalized root mean square error, and maximum group delay error. At the same time, its performance in amplitude frequency response and group delay response was also good. This indicates that the design ideas and methods are effective and can provide better performance guarantees for the application of VFD filters.

With the continuous development of signal processing technology and the expansion of application fields, future research needs to continue to delve into the optimization design methods of VFD filters, explore more efficient and accurate algorithms and models to meet the needs of different application scenarios.

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Informed consent/Patient consent

Not applicable.

Trial registration number/date

Not applicable.

Ethical Statement

There are no conflicts of interest in this study.

Data availability statement

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Author contribution

All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Hai Feng Zheng. The first draft of the manuscript was written by Li Ying Han, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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